The New Classical Approach

Adapted by Dr Camilo Calderon

Table of contents

1 Introduction

The New Classical approach extends monetarist skepticism about policy fine-tuning by sharpening the mechanism through which policy affects real outcomes. The key shift is from short-run money illusion under adaptive expectations (Friedman's framework) to model-consistent forecasting under rational expectations, which removes systematic policy surprises and hence eliminates systematic real effects of monetary policy.

1.1 Expectations and the rational expectations revolution

1.1.1 Adaptive expectations

Adaptive expectations operate as a trial-and-error forecasting rule where agents update forecasts using past observed outcomes. A generic representation is:

Price expectations formed at time $t-1$ for time t: $E_{t-1}P_t = f(P_{t-1}, P_{t-2}, \ldots)$

A common special case (error-correction rule) is:

$$\pi_t^e = \pi_{t-1}^e + \lambda(\pi_{t-1} - \pi_{t-1}^e),$$

with $0 < \lambda \leq 1$. The key point is that adaptive expectations permit systematic forecast errors during regime changes, creating temporary money illusion and thus temporary real effects.

1.1.2 Rational expectations

Rational expectations are defined using three criteria: (i) forecasts are formed using the true model (agents know the economy's structure as well as policymakers do), (ii) forecasts condition on the best available information at the time, and (iii) forecast errors are purely random (white noise) relative to the information set. Systematic errors are eliminated.

In its classic formulation, the rational expectations hypothesis asserts that expectations are "essentially the same as the predictions of the relevant economic theory," and that the economy does not waste information.

Model-consistent forecasting is written as:

$$E_{t-1}P_t = E(P_t \mid I_{t-1}),$$

where I_{t-1} is the information set available at time $t-1$.

A core implication is that if workers have access to statistically unbiased forecasts, predictable policy loses its ability to manipulate perceived real wages and hence loses its ability to systematically shift real output, even in the short run.

2 The New Classical model of output and prices

This section reconstructs the core New Classical model, highlighting what is assumed, what is solved for, and what the key parameter κ means.

2.1 Lucas aggregate supply and aggregate demand

The New Classical macro model consists of two equations: a Lucas supply relation (LAS) and an aggregate demand (AD) relation derived from a quantity-theory style spending equation.

Lucas aggregate supply (LAS). Output supplied deviates from the natural (full-employment) level only when the price level differs from what was expected:

$$Y_t^s - Y^* = \kappa(P_t - E_{t-1}P_t).$$

Lucas's 1972 analysis derives this relationship in a framework that excludes money illusion: markets clear, agents optimise, and expectations are formed optimally.

Interpretation: - If $P_t > E_{t-1}P_t$, firms perceive a favourable relative-price signal and supply more output, so $Y_t^s > Y^*$. - If $P_t = E_{t-1}P_t$, output is at potential: $Y_t^s = Y^*$. - If $P_t < E_{t-1}P_t$, output is below potential: $Y_t^s < Y^*$.

The parameter κ is the responsiveness of supply to a price surprise: $\partial Y_t/\partial(P_t - E_{t-1}P_t) = \kappa$.

Aggregate demand (AD). In log form, AD is:

$$Y_t^d = \gamma + M_t - P_t,$$

where M_t is the log of the nominal money supply and γ captures exogenous demand shifters.

A key assumption is that agents know the structural parameters γ, κ, and the natural output level Y^*.

2.2 Solving the model and the role of κ

Equilibrium requires $Y_t^s = Y_t^d = Y_t$. Combining LAS and AD yields the central results:

1. Under rational expectations, agents understand LAS, so expected output is the natural level:
$$E_{t-1}Y_t = Y^*.$$
2. Output deviations are proportional to the unexpected component of money:
$$Y_t - Y^* = \frac{\kappa}{1+\kappa}(M_t - E_{t-1}M_t).$$

3. Price surprises are also proportional to monetary surprises:

$$P_t - E_{t-1}P_t = \frac{1}{1+\kappa}(M_t - E_{t-1}M_t).$$

Interpretation: A larger κ means output responds more and prices respond less to a given monetary surprise; a smaller κ means the opposite. Geometrically (with P on the vertical axis and Y on the horizontal axis), the slope of the Lucas aggregate supply curve is proportional to $1/\kappa$.

3 Monetary policy implications

3.1 The monetary policy ineffectiveness proposition

The central result is the **monetary policy ineffectiveness proposition**: with rational expectations and market clearing, *systematic and predictable* monetary policy has **no effect on real variables**, even in the short run. The reason is not "policy doesn't matter," but that predictable policy is incorporated into expectations, so it moves nominal variables (prices) without creating systematic price surprises.

Formally, if the public can forecast money supply, then $M_t = E_{t-1}M_t$, so the monetary surprise term is zero and output remains at Y^*:

$$\text{If } M_t = E_{t-1}M_t, \text{ then } Y_t - Y^* = 0.$$

3.2 Systematic rules versus random disturbances

A sharp distinction exists between (i) predictable rule-like policy and (ii) random monetary disturbances.

A stylised money supply process is:

$$M_t = (1+g_M)M_{t-1} + u_t,$$

where u_t is white noise with $E_{t-1}u_t = 0$. Then:

$$E_{t-1}M_t = (1+g_M)M_{t-1},$$

so $M_t - E_{t-1}M_t = u_t$.

Substituting into the reduced form gives:

$$Y_t - Y^* = \frac{\kappa}{1+\kappa}u_t.$$

Only the random and unpredictable component of money affects output. Predictable money growth affects the price level, not real activity.

3.3 Why "be unpredictable" is not a recommendation

The model should not be misread as an argument for deliberate randomness. Two reasons are evident:

1. Random policies produce recessions as well as booms: negative u_t is as likely as positive u_t.
2. High volatility changes behaviour: if agents expect frequent noise, they discount price signals more heavily, making LAS steeper (reducing the output response and increasing the price response).

This is exactly the logic behind modern central-bank emphasis on transparency and systematic frameworks: policy becomes more effective when expectations are well-anchored and communication reduces unnecessary uncertainty.

4 Exceptions and bridges to other approaches

The analysis includes two important exceptions—cases where monetary policy can systematically stabilise real activity without violating rational expectations, because the key assumption is weakened (informational symmetry or full nominal flexibility).

4.1 Asymmetric information and real shocks

If fluctuations are driven by productivity shocks and the central bank receives information enabling it to forecast a negative productivity shock before the private sector does, it can implement expansionary policy in advance to stabilise output and employment (at the cost of a higher price level).

This argument is consistent with the New Classical framework because the stabilising action is effectively unexpected from the public's perspective: the central bank has an informational advantage, so policy can create a gap between the expected and actual price level.

This logic is compatible with real-business-cycle reasoning, where technology/productivity shocks are central drivers of cycles.

4.2 Long-term nominal wage contracts and counter-cyclical stabilisation

In many real economies, nominal wages are fixed by multi-year contracts, often negotiated by unions. Even if everyone can predict an aggregate demand shock and anticipate the central bank's response, monetary policy can have real effects because some nominal wages cannot adjust immediately. It is the rigidity itself that creates scope for stabilisation.

This is consistent with Fischer's formal analysis of long-term contracts under rational expectations, where (in the presence of wage rigidity) the design of monetary rules interacts with the propagation of shocks—even when expectations are rational.

Under symmetric information (the central bank has no informational advantage), the stabilisation logic applies most naturally to aggregate demand shocks; supply-side stabilisation requires informational asymmetry.

4.3 Lucas critique and credibility

A major New Classical contribution is the **Lucas critique**: estimates of policy effects from historical macro relationships are unreliable if private decision rules shift when the policy regime shifts. This warning motivates structural (microfounded) models for policy evaluation.

If agents form expectations rationally, then **commitment and credibility** shape outcomes through expectations. This is why rules, clear strategy statements, and transparent communication are policy tools.

In the time-inconsistency literature, policy is a strategic interaction with rational agents: discretionary optimisation can be self-defeating because promises today are not optimal tomorrow (once expectations are set). This perspective is central to rules-versus-discretion analysis.

5 Modern evidence and real-world examples

5.1 Measuring policy surprises

In the New Classical model, the disturbance term is $M_t - E_{t-1}M_t$. Modern empirical work isolates *surprise components* of policy announcements using high-frequency financial market data.

A recent IMF working paper constructs a dataset of monetary policy shocks for 21 advanced economies and 8 emerging markets (2000–2022) using daily changes in interest rate swap rates around central bank announcements to identify unexpected shocks to the policy path. This maps directly onto the model's object: an unexpected policy shock is the empirical analogue of u_t in the money-growth process.

5.2 Credibility and anchored expectations

New Classical analysis makes credibility and expectations central. Modern central banks explicitly frame policy as expectations management.

The Federal Reserve explains that publishing its "Statement on Longer-Run Goals and Monetary Policy Strategy" helps the public understand the policy framework, which matters for transparency, accountability, and effectiveness.

The ECB's 2021 strategy statement emphasises a symmetric 2% medium-term inflation aim and explicitly notes that the effective lower bound can require "especially forceful or persistent" measures, potentially including a transitory period of inflation moderately above target.

IMF research provides evidence that better-anchored expectations are associated with less persistent inflation responses to shocks, reinforcing the New Classical emphasis on credibility and expectations formation.

5.3 Policy regime episodes

The Bank of Japan reference timeline documents major unconventional regimes from the late 1990s onward (ZIRP from 1999, QE from 2001, and later phases). It explicitly lists policy duration effect (forward guidance) as part of ZIRP—an expectations-management concept compatible with New Classical thinking.

Japan's 2024 framework change statement explains that QQE with yield curve control and negative rates were judged to have fulfilled their roles, with a shift back to guiding the short-term interest rate as the primary tool.

For the euro area, the ECB's June 2014 press release introducing a negative deposit facility rate provides a canonical policy regime expansion example: the ECB cut the deposit facility rate to -0.10% effective June 11, 2014.

During COVID-19, emergency actions show how central banks reacted to extraordinary shocks with large-scale measures that were partly about market functioning and partly about macro stabilisation.

6 Figures

Figure 1. Lucas aggregate supply and equilibrium at potential output. Output deviates from Y^* only when the price level differs from its rational expectation. If expectations are correct, output is at potential.

```
Warning: package 'ggplot2' was built under R version 4.4.3

Warning in geom_point(aes(x = Y_star, y = E_P), color = "black", size = 3): All aesthetics ha
i Please consider using `annotate()` or provide this layer with data containing
  a single row.
```

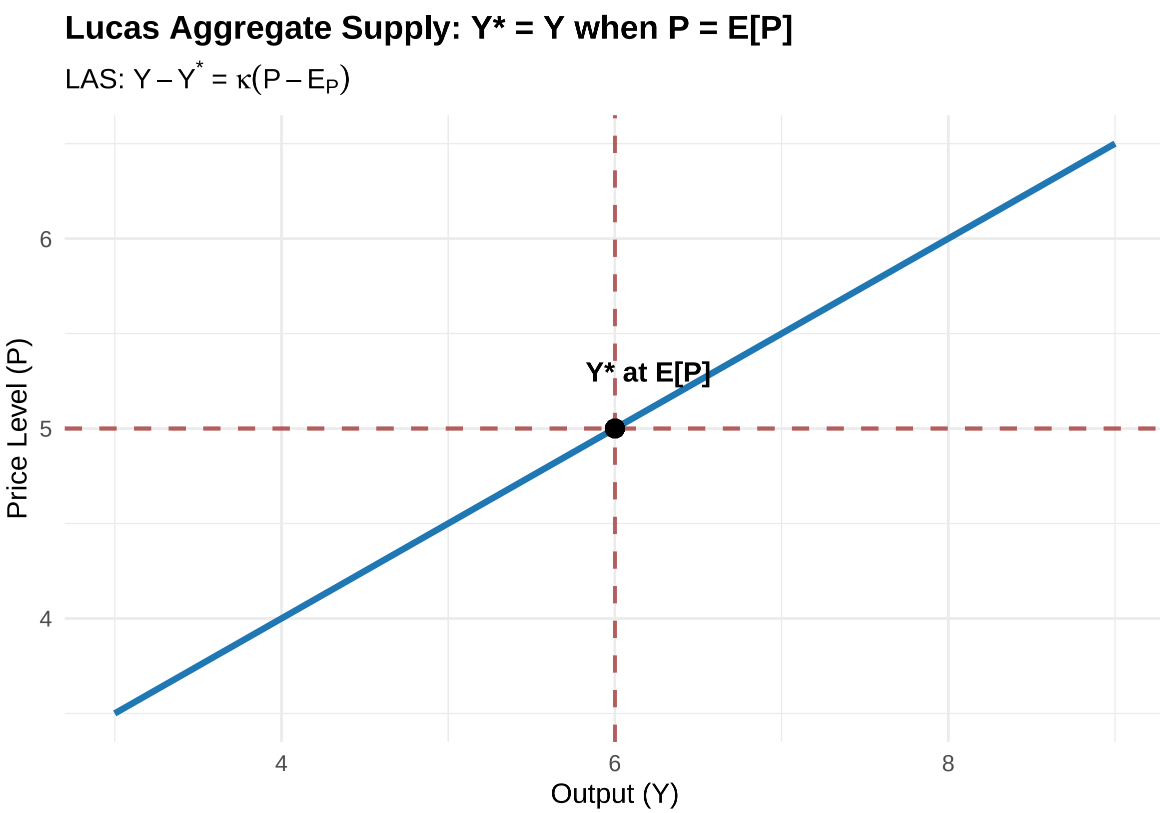

Figure 1: Lucas Aggregate Supply: output deviates from Y* only when P differs from E[P].

Figure 2. Predictable versus unpredictable money: shifts in AD and LAS. A predictable rise in money shifts AD and also shifts LAS upward because expected prices rise; equilibrium stays at Y^*. An unpredictable rise in money shifts AD without immediately shifting expectations; equilibrium moves temporarily along LAS with Y above Y^*.

```
Warning: package 'gridExtra' was built under R version 4.4.3
```

```
Warning in geom_point(aes(x = Y_recession, y = P_recession), color = "red", : All aesthetics
i Please consider using `annotate()` or provide this layer with data containing
  a single row.
```

```
Warning in geom_point(aes(x = Y_stabilised, y = P_stabilised), color = "darkgreen", : All aes
i Please consider using `annotate()` or provide this layer with data containing
  a single row.
```

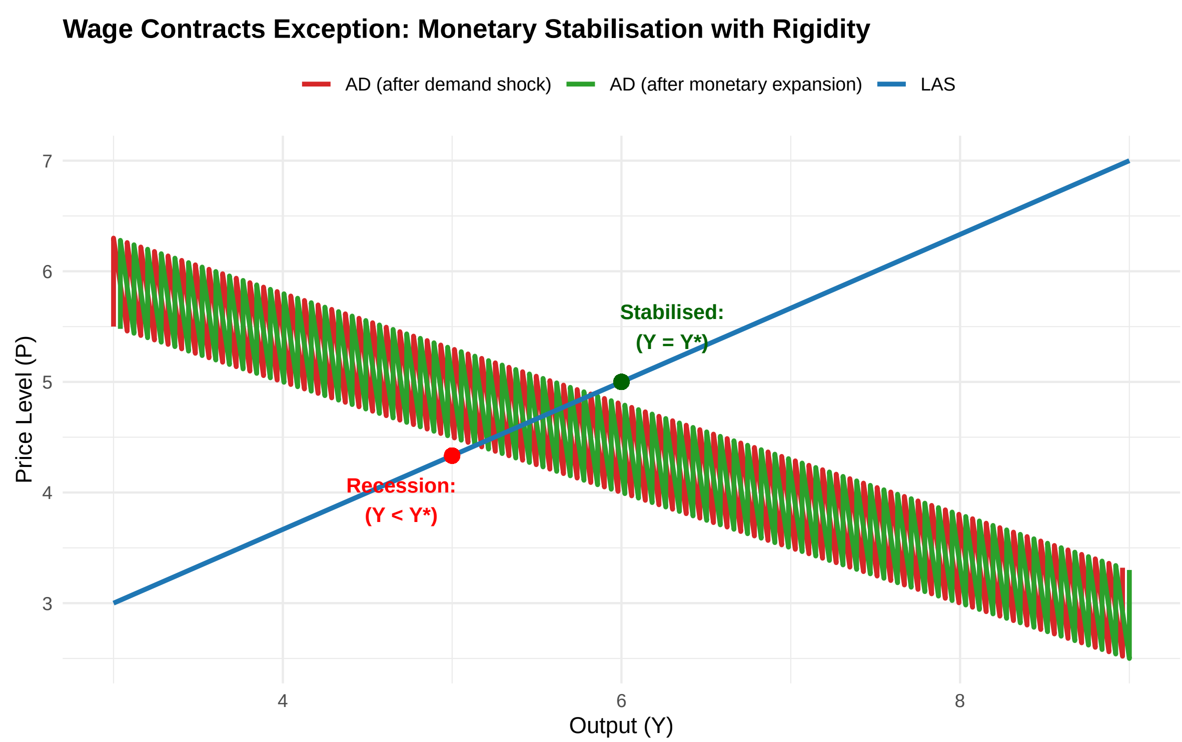

Figure 2: Comparing predictable policy (Y stays at Y*) vs unpredictable policy (Y temporarily rises).

7 Problem questions

1. In the LAS equation $Y_t - Y^* = \kappa(P_t - E_{t-1}P_t)$, interpret every term economically. Why does the model predict $E_{t-1}Y_t = Y^*$ under rational expectations?
2. Show that equilibrium output deviations are $Y_t - Y^* = (\kappa/(1+\kappa))(M_t - E_{t-1}M_t)$. What happens as $\kappa \to 0$? What happens as $\kappa \to \infty$?
3. Explain why "random policy can move output" is not an argument for deliberate policy randomness. Explain using both (i) the sign symmetry of shocks and (ii) the "boy who cried wolf" mechanism.
4. Describe the asymmetric-information exception: what must be true about the central bank's information set relative to the private sector's for systematic stabilisation to operate under rational expectations?
5. Explain the wage-contract critique: why can anticipated monetary policy offset demand shocks when only part of the economy's wages are flexible?
6. Using high-frequency monetary policy shock data from modern empirical work, explain how researchers operationalise "policy surprises." Why is this relevant to the New Classical policy-ineffectiveness result?

8 References

Fischer, S. (1977). Long-term contracts, rational expectations, and the optimal money supply rule. *Journal of Political Economy*, 85(1), 191–205.

International Monetary Fund. (2024). A new dataset of high-frequency monetary policy shocks (Working Paper 2024/224).

Lucas, R. E., Jr. (1972). Expectations and the neutrality of money. *Journal of Economic Theory*, 4(2), 103–124.

Muth, J. F. (1961). Rational expectations and the theory of price movements. *Econometrica*, 29(3), 315–335.

Prescott, E. C. (1986). Theory ahead of business cycle measurement. *Quarterly Review* / Federal Reserve Bank of Minneapolis (Staff Report 102).

Sargent, T. J., & Wallace, N. (1976). Rational expectations and the theory of economic policy. *Journal of Monetary Economics*, 2(2), 169–183.

European Central Bank. (2014). ECB introduces a negative deposit facility interest rate (Press release, 5 June).

European Central Bank. (2021). The ECB's monetary policy strategy statement.

Bank of England. (2020). Bank Rate reduced to 0.1% and asset purchases increased by £200bn (Monetary Policy Summary, 19 March).

Bank of Japan. (2024). Changes in the Monetary Policy Framework (19 March).

Bank of Japan. (n.d.). Unconventional Monetary Policy Measures from the Late 1990s.